Year 2 Pure Mathematics

Chapter 1

Algebraic methods

Practice Booklet

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1.1 Proof by contradiction

Practice Book, printed page 1.

1. Write down the negation of each statement.

a. There are infinitely many prime numbers.

b. If n^2 is even, then n must be even.

c. If pq is odd, then at least one of p and q is odd.

Hint

A statement that asserts the falsehood of another statement is called the **negation** of the statement.

2. Prove by contradiction that there is no greatest even number.

Hint

You prove a statement by contradiction by assuming it is not true. Then use logical steps to show your assumption leads to something impossible. You can then conclude that your assumption was incorrect, and the original statement was true.

3. Prove by contradiction that if a and b are integers, then $a^2 - 4b - 7 \neq 0$.

Hint

Start by assuming that there exist integers a and b such that $a^2 - 4b - 7 = 0$. Rearrange and use the fact that any odd number can be written in the form $2n + 1$, where n is an integer.

4. Prove by contradiction that there is no smallest positive rational number.

Hint

A **rational number** can be written in the form $\frac{a}{b}$, where a and b are integers. An **irrational number** cannot be written in the form $\frac{a}{b}$, where a and b are integers. The set of rational numbers can be written as $\mathbb{Q}$.

5. E/P Prove by contradiction that if n^3 is odd, then n must be odd. (3 marks)

6. E/P Prove by contradiction that there are no non-zero integer solutions to the equation $x^2 - y^2 = 1$. (4 marks)

7. E/P Prove that $\sqrt{5}$ is irrational. (6 marks)

8. E/P Prove by contradiction that the difference between any rational number and any irrational number is irrational. (6 marks)

1.2 Algebraic fractions

Practice Book, printed page 1.

1. Simplify:

a. $(x-3) \times \dfrac{1}{x^2-9}$

b. $\dfrac{x^2-1}{3} \times \dfrac{1}{x^2+2x+1}$

c. $\dfrac{x^2+5x}{y-2} \times \dfrac{y^2-2y}{x^2}$

Hint

To multiply algebraic fractions, first factorise the numerators and denominators where possible. Then cancel any common factors, and multiply the numerators and multiply the denominators.

2. Simplify:

a. $\dfrac{x}{x+2} \div \dfrac{x^3}{x^2+x-2}$

b. $\dfrac{9x^2-16}{5x-10} \div \dfrac{3x-4}{10}$

c. $\dfrac{2x^2+3xy-2y^2}{5xy} \div \dfrac{x^2+2xy}{10x^2}$

Hint

To divide two algebraic fractions, multiply the first fraction by the reciprocal of the second fraction.

3. Express as a single fraction in its simplest form:

a. $\dfrac{5}{x+3} + \dfrac{2}{x-1}$

b. $\dfrac{4}{2(x-3)} + \dfrac{5}{3(x+1)}$

c. $\dfrac{2x}{x-5} - \dfrac{3x}{x+5}$

d. $\dfrac{x+1}{2x-3} - \dfrac{2x}{x+2}$

Hint

To add or subtract two fractions, find a common denominator.

4. Express as a single fraction in its simplest form:

a. $\dfrac{5x}{4x^2-9} + \dfrac{3}{2x-3}$

b. $\dfrac{2}{x^2+x-12} + \dfrac{1}{x^2-5x+6}$

c. $\dfrac{5}{4x^2+4x+1} - \dfrac{3}{4x^2-1}$

d. $\dfrac{x-1}{x^2+3x+2} - \dfrac{x-2}{x^2-2x-3}$

Hint

You may need to factorise the denominators to find the lowest common multiple.

5. E/P Simplify $\dfrac{x^2+4x+4}{y^2-6y+9} \div \dfrac{x^2-4}{y^2-9}$ (4 marks)

6. E/P Simplify $\dfrac{x^2-2x-15}{2x^2-12} \times \dfrac{x^3-6x^2}{x^2-5x-24}$ (4 marks)

7. E/P Express $\dfrac{2x^2-5x}{25x^2-1}+\dfrac{3x}{5x-1}$ as a fraction in its simplest form. (4 marks)

8. E/P Express $\dfrac{3x-2}{2x^2-5x-3}-\dfrac{5}{2x+1}$ as a fraction in its simplest form. (4 marks)

9. E/P $\mathrm{f}(x)=x+\dfrac{5}{x+3}+\dfrac{40}{x^2-2x-15}$, $x\in\mathbb{R}$, $x\neq -3$, $x\neq 5$

a. Show that $\mathrm{f}(x)=\dfrac{x^3-2x^2-10x+15}{(x+3)(x-5)}$ (4 marks)

b. Hence show that $\mathrm{f}(x)$ can be further simplified to give $\mathrm{f}(x)=\dfrac{x^2-5x+5}{x-5}$ (4 marks)

1.3 Partial fractions

Practice Book, printed page 2.

1. Express in partial fractions:

a. $\dfrac{8x-1}{(x+1)(x-2)}$

b. $\dfrac{2x+13}{(2x-1)(x+3)}$

c. $\dfrac{7-11x}{(3x-1)(2x+1)}$

Hint

For part a, write

$$\frac{8x-1}{(x+1)(x-2)}\equiv\frac{A}{x+1}+\frac{B}{x-2}$$

Then find A and B by rearranging and substituting suitable values of x.

2. Express in partial fractions:

a. $\dfrac{7x-15}{x^2-5x}$

b. $\dfrac{3(x+7)}{x^2-9}$

c. $\dfrac{9x-1}{2x^2-9x-5}$

Hint

Factorise each denominator to work out the denominators for the partial fractions.

3. Express in partial fractions:

a. $\dfrac{6x^2-43x+50}{x(x-2)(x-5)}$

b. $\dfrac{4x^2+11x+9}{(x-1)(x+2)(x+3)}$

c. $\dfrac{5x^2-22x+6}{x(x-3)(2x-1)}$

Hint

For part a, write

$$\frac{6x^2-43x+50}{x(x-2)(x-5)}\equiv\frac{A}{x}+\frac{B}{x-2}+\frac{C}{x-5}$$

You can also find A, B and C by multiplying across and comparing coefficients. Often a combination of substitution and comparing coefficients is the most efficient way to find the numerators.

4. Given that $\dfrac{1+15x-10x^2}{(x-2)(1-2x)} \equiv A + \dfrac{B}{x-2} + \dfrac{C}{1-2x}$, find the values of the constants A, B and C.

Hint

Start by multiplying both sides by $(x-2)(1-2x)$ to remove the fractions.

5. E/P Given that $\dfrac{4}{(x+1)(3x-2)} \equiv \dfrac{A}{x+1} + \dfrac{B}{3x-2}$, find the values of the constants A and B. (3 marks)

6. E/P Express $\dfrac{5x-3}{(2x+3)(x+2)}$ in partial fractions. (3 marks)

7. E/P Express $\dfrac{3}{9-y^2}$ in partial fractions. (3 marks)

8. E/P Given that $\dfrac{33x-x^2-44}{(x-1)(x+5)(2x-3)} \equiv \dfrac{A}{x-1} + \dfrac{B}{x+5} + \dfrac{C}{2x-3}$, find the values of the constants A, B and C. (4 marks)

9. E/P Given that $\dfrac{2x^2-11}{(x+1)(x-2)} \equiv A + \dfrac{B}{x+1} + \dfrac{C}{x-2}$, find the values of the constants A, B and C. (4 marks)

1.4 Repeated factors

Practice Book, printed page 3.

1. $\mathrm{f}(x) = \dfrac{5x^2-5x+2}{x^2(x-2)}$, $x \neq 0$, $x \neq 2$. Given that $\mathrm{f}(x)$ can be expressed in the form $\dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x-2}$, find the values of the constants A, B and C.

Hint

x is a repeated factor, so you need separate denominators of x and x^2 in the expanded partial fraction.

2. $\mathrm{g}(x) = \dfrac{8x^2-x+3}{(x+1)^2(2x-1)}$, $x \neq -1$, $x \neq \dfrac{1}{2}$

Given that $\mathrm{g}(x)$ can be expressed in the form $\dfrac{A}{x+1} + \dfrac{B}{(x+1)^2} + \dfrac{C}{2x-1}$, find the values of the constants A, B and C.

Hint

$(x+1)$ is a repeated linear factor, so you need denominators of $(x+1)$ and $(x+1)^2$.

3. E/P $\dfrac{4x-1}{(2x+1)^2} \equiv \dfrac{A}{2x+1} + \dfrac{B}{(2x+1)^2}$. Find the values of the constants A and B. (3 marks)

4. E/P Given that $\dfrac{x^2+7x+32}{(x-2)(x+3)^2} \equiv \dfrac{A}{x-2} + \dfrac{B}{x+3} + \dfrac{C}{(x+3)^2}$, find the values of the constants A, B and C. (4 marks)

5. E/P Express $\dfrac{6x^2-13x+15}{2x^3-3x^2}$ in the form $\dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{2x-3}$, where A, B and C are constants to be found. (4 marks)

6. E/P Express $\dfrac{x^2-5x+16}{x^3-4x^2+4x}$ as a sum of partial fractions. (6 marks)

1.5 Algebraic division

Practice Book, printed page 4.

1. Use algebraic division to show that

$$\frac{x^2 - 5x + 7}{x - 3} \equiv x - 2 + \frac{1}{x - 3}$$

Hint

This is an improper algebraic fraction because the degree of the numerator is greater than or equal to the degree of the denominator. You can divide the numerator by the denominator to obtain a polynomial and a remainder.

2.

a. Find the remainder when $x^2 - 5x + 9$ is divided by $x + 1$.

b. Hence, or otherwise, write $\frac{x^2 - 5x + 9}{x + 1}$ in the form $Ax + B + \frac{C}{x + 1}$, where A, B and C are constants to be found.

Hint

For part b, you can use the relationship $\mathrm{F}(x) = \mathrm{Q}(x) \times \text{divisor} + \text{remainder}$ to write

$$\frac{\mathrm{F}(x)}{\text{divisor}} = \mathrm{Q}(x) + \frac{\text{remainder}}{\text{divisor}}$$

3. Given that $\frac{x^3 - x^2 - 9}{x + 3} \equiv Ax^2 + Bx + C + \frac{D}{x + 3}$, find the values of the constants A, B, C and D.

Hint

You could use algebraic long division, or you could multiply both sides by $(x + 3)$ and use substitution and comparing coefficients.

4. Given that

$$x^3 - 2x^2 + 5 \equiv (Ax^2 + Bx + C)(x - 4) + D$$

find the values of the constants A, B, C and D.

Hint

You can compare coefficients when the identity is written in this form.

5. E/P $\frac{18x^2 + 22x - 7}{(x + 2)(3x - 1)} \equiv A + \frac{B}{x + 2} + \frac{C}{3x - 1}$. Find the values of the constants A, B and C.(4 marks)

6. E/P Given that $\frac{2x^3 - 4x^2 + 5x - 1}{x - 3} \equiv Ax^2 + Bx + C + \frac{D}{x - 3}$, find the values of the constants A, B, C and D. (4 marks)

7. E/P Given that $\frac{x^4 - 3x^2 + 5}{x^2 + 2} \equiv Ax^2 + B + \frac{C}{x^2 + 2}$, find the values of the constants A, B and C. (4 marks)

8. E/P Given that $\frac{4x^2 - 5x - 3}{(x + 1)(2x - 1)} \equiv A + \frac{B}{x + 1} + \frac{C}{2x - 1}$, find the values of the constants A, B and C. (4 marks)

9. E/P $3x^4 - 5x^3 + 6x^2 - 12x + 5 \equiv (Ax^2 + Bx + C)(x^2 + 2) + Dx + E$
Find the values of the constants A, B, C, D and E. (5 marks)

Problem solving: Set A

Bronze

a. Simplify $\dfrac{3x^2+x-2}{x^2-1}$ (3 marks)

b. Hence, or otherwise, express $\dfrac{3x^2+x-2}{x^2-1} - \dfrac{1}{x^2-x}$ as a single fraction in its simplest form. (3 marks)

Silver

Express $\dfrac{2x^2-3x}{2x^2+x-6} - \dfrac{6}{x^2+x-2}$ as a single fraction in its simplest form. (7 marks)

Gold

Express $\dfrac{2x^3+3x^2-5x-6}{3x^2+5x+2}$ in the form $Ax+B+\dfrac{C}{Dx+E}$, where A, B, C, D and E are constants to be found. (9 marks)

Problem solving: Set B

Bronze

Express $\dfrac{39x^2-49x+15}{(3x-2)^2(1-x)}$ as a sum of partial fractions. (4 marks)

Silver

Show that $\dfrac{3x^2-2x+4}{x^2-x-6}$ can be written in the form $A+\dfrac{B}{x-3}+\dfrac{C}{x+2}$ and find the values of the constants A, B and C. (5 marks)

Gold

$\mathrm{f}(x) = \dfrac{3x^2+3}{2x^3-5x^2-4x+3}$

The graph of $y=\mathrm{f}(x)$ has a vertical asymptote with equation $x=-1$. Use this information to express $\mathrm{f}(x)$ as the sum of three fractions with linear denominators. (8 marks)

Now try this ⟶ Exam question bank Q5, Q9, Q49, Q62, Q69, Q75, Q78

Selected exam question bank questions

Question 5

Given that $\dfrac{2x^4+3x^2-5x+2}{x^2-4} = ax^2+bx+c+\dfrac{d}{x+2}+\dfrac{e}{x-2}$, find the values of the constants a, b, c, d and e. (6 marks)

Question 9

Express $\dfrac{7x^2+12x}{(x+1)^2(x+2)}$ in partial fractions. (5 marks)

Question 49

$$\mathrm{f}(x) = \frac{12x^2+20x+12}{(3x+1)(x+1)(x+2)} = \frac{A}{3x+1}+\frac{B}{x+1}+\frac{C}{x+2}$$

a. Find the values of the constants A, B and C. (3 marks)

b. Hence find the exact value of $\displaystyle\int_0^2 \mathrm{f}(x)\,\mathrm{d}x$, giving your answer in the form $\ln\dfrac{a}{b}$ where $\dfrac{a}{b}$ is a fraction in its simplest form. (5 marks)

Question 62

a. "If p and q are irrational numbers, then pq is also an irrational number."
Disprove this statement using a counter example. (2 marks)

b. Prove that $\sqrt{2}$ is an irrational number. (6 marks)

Question 69

The function f is defined by $\mathrm{f}: x \mapsto \dfrac{2(x-1)}{x^2-2x-8}-\dfrac{1}{x-4}$, $x>4$

a. Show that $\mathrm{f}(x) = \dfrac{1}{x+2}$, $x>4$. (4 marks)

b. Find the range of $\mathrm{f}(x)$. (2 marks)

c. Find $\mathrm{f}^{-1}(x)$. State the domain of this inverse function. (3 marks)

The function g is defined by $\mathrm{g}: x \mapsto 2x^2+1$, $x \in \mathbb{R}$

d. Solve $\mathrm{fg}(x) = \dfrac{1}{43}$ (3 marks)

Question 75

a. Express $\dfrac{5x^2-4x}{5x^2+6x-8}-\dfrac{10}{x^2-x-6}$ as a single fraction in its simplest form. (7 marks)

b. Hence write $\dfrac{5x^2-4x}{5x^2+6x-8}-\dfrac{10}{x^2-x-6}$ in the form $A+\dfrac{B}{x-3}$, where A and B are constants to be found. (2 marks)

$$\mathrm{f}(x) = \frac{5x^2-4x}{5x^2+6x-8}-\frac{10}{x^2-x-6},\ x \neq 3$$

c. Use your answer to part b to prove that $\mathrm{f}(x)$ is an increasing function for all $x>3$. (3 marks)

Question 78

$$\frac{3x^2 - x + 2}{(x+1)(x-2)} \equiv A + \frac{B}{x+1} + \frac{C}{x-2}$$

a. Find the values of the constants A, B and C. (4 marks)

b. Hence, or otherwise, expand $\frac{3x^2 - x + 2}{(x+1)(x-2)}$ in ascending powers of x, up to and including the term in x^2. Give each coefficient as a simplified fraction. (7 marks)

c. State the set of values for which the expansion in part b is valid. (1 mark)

Answers

1.1 Proof by contradiction

1.

a. There are a finite number of prime numbers.

b. There exists an n such that n^2 is even but n is odd.

c. There exist p and q such that pq is odd and p and q are both even.

2. Assume that there is a greatest even number, n. Then $n+2 > n$ and is also even. This contradicts the assumption, so there is no greatest even number.

3. Assume that $a^2 - 4b - 7 = 0$, where a and b are integers. So $a^2 = 4b + 7$. Since $4b + 7$ is odd, a^2 is odd and so a is odd and can be written as $2n + 1$ where n is an integer.
Then $a^2 = (2n+1)^2 = 4b + 7 \Rightarrow 4n^2 + 4n + 1 = 4b + 7$
Rearranging, $4n^2 + 4n - 4b = 6 \Rightarrow 4(n^2 + n - b) = 6$
So $n^2 + n - b = \dfrac{3}{2}$, which is not an integer, but as n and b are integers, $n^2 + n - b$ must also be an integer. This contradiction implies that $a^2 - 4b - 7$ cannot be equal to 0.

4. Assume that there is a smallest positive rational number, n. Then $n > 0$ and $n = \dfrac{a}{b}$ where a and b are integers and $b \neq 0$.
Then $\dfrac{n}{2} = \dfrac{a}{2b}$, and since b is an integer, $2b$ is also an integer.
This means that $\dfrac{n}{2}$ is rational and as $\dfrac{n}{2} < n$ this is a contradiction, which implies there is no smallest positive rational number.

5. Assume that n is even. Then $n = 2p$ where p is an integer. $n^3 = (2p)^3 = 8p^3$ which is even. Thus if n^3 is odd, then n must also be odd.

6. Assume that there exist non-zero integers x and y such that $x^2 - y^2 = 1$. Then $(x+y)(x-y) = 1$. The only factor of 1 is 1. There are 2 possibilities using $1 \times 1 = 1$ and $-1 \times -1 = 1$:

$$x + y = 1 \text{ and } x - y = 1 \qquad (1)$$

$$x + y = -1 \text{ and } x - y = -1 \qquad (2)$$

(1) gives $x = 1$ and $y = 0$ and (2) gives $x = -1$ and $y = 0$.
$y = 0$ is a solution in both cases, so there are no non-zero integer solutions to the equation $x^2 - y^2 = 1$.

7. Assume that $\sqrt{5}$ is rational. Then $\sqrt{5} = \dfrac{a}{b}$, where a and b are integers, $b \neq 0$ and the highest common factor of a and b is 1. Then $a = \sqrt{5}b \Rightarrow a^2 = 5b^2$.
So a^2 is divisible by 5 and it must contain a (repeated) factor of 5, and so a is divisible by 5. So $a = 5p$, where p is an integer.
$\Rightarrow (5p)^2 = 5b^2 \Rightarrow b^2 = 5p^2$. So b^2 is divisible by 5 and so b is divisible by 5. As a and b are both divisible by 5, then the highest common factor of a and b is not 1. This contradiction implies that $\sqrt{5}$ is irrational.

8. Assume that p is rational, q is irrational, and that $p - q$ is rational. Then there exist integers a, b, c and d such that

$$p = \frac{a}{b} \text{ and } p - q = \frac{c}{d},\ b, d \neq 0.$$

So $\dfrac{a}{b} - q = \dfrac{c}{d} \Rightarrow q = \dfrac{a}{b} - \dfrac{c}{d} = \dfrac{ad - bc}{bd}$
But $ad - bc$ is an integer as a, b, c and d are all integers and $bd \neq 0$, which means that q is rational. This is a contradiction. So the difference between any rational number and any irrational number is irrational.

1.2 Algebraic fractions

1.

a. $\dfrac{1}{x+3}$

b. $\dfrac{x-1}{3(x+1)}$

c. $\dfrac{y(x+5)}{x}$

2.

a. $\dfrac{x-1}{x^2}$

b. $\dfrac{2(3x+4)}{x-2}$

c. $\dfrac{2(2x-y)}{y}$

3.

a. $\dfrac{7x+1}{(x+3)(x-1)}$

b. $\dfrac{11x-9}{3(x-3)(x+1)}$

c. $\dfrac{x(25-x)}{(x+5)(x-5)}$

d. $\dfrac{2+9x-3x^2}{(2x-3)(x+2)}$

4.

a. $\dfrac{11x+9}{(2x+3)(2x-3)}$

b. $\dfrac{3x}{(x+4)(x-3)(x-2)}$

c. $\dfrac{4x-8}{(2x+1)^2(2x-1)}$

d. $\dfrac{7-4x}{(x+2)(x+1)(x-3)}$

5. $\dfrac{(x+2)(y+3)}{(x-2)(y-3)}$

6. $\dfrac{x^2(x-5)(x-6)}{2(x^2-6)(x-8)}$

7. $\dfrac{x(17x-2)}{(5x+1)(5x-1)}$

8. $\dfrac{13-2x}{(2x+1)(x-3)}$

9.

a. $\mathrm{f}(x) = \dfrac{x(x+3)(x-5)+5(x-5)+40}{(x+3)(x-5)} = \dfrac{x^3-2x^2-10x+15}{(x+3)(x-5)}$

b. Using the factor theorem, $(x+3)$ is a factor of

$$x^3-2x^2-10x+15 = (x+3)(x^2-5x+5)$$

So $\mathrm{f}(x) = \dfrac{x^2-5x+5}{x-5}$

1.3 Partial fractions

1.

a. $\dfrac{3}{x+1} + \dfrac{5}{x-2}$

b. $\dfrac{4}{2x-1} - \dfrac{1}{x+3}$

c. $\dfrac{2}{3x-1} - \dfrac{5}{2x+1}$

2.

a. $\dfrac{3}{x} + \dfrac{4}{x-5}$

b. $\dfrac{5}{x-3} - \dfrac{2}{x+3}$

c. $\dfrac{1}{2x+1} + \dfrac{4}{x-5}$

3.

a. $\dfrac{5}{x} + \dfrac{2}{x-2} - \dfrac{1}{x-5}$

b. $\dfrac{2}{x-1} - \dfrac{1}{x+2} + \dfrac{3}{x+3}$

c. $-\dfrac{2}{x} - \dfrac{1}{x-3} + \dfrac{3}{2x-1}$

4. $A = 5$, $B = 3$, $C = -4$

5. $A = -\dfrac{4}{5}$, $B = \dfrac{12}{5}$

6. $\dfrac{13}{x+2} - \dfrac{21}{2x+3}$

7. $\dfrac{1}{2(3-y)} + \dfrac{1}{2(3+y)}$

8. $A = 2$, $B = -3$, $C = 1$

9. $A = 2$, $B = 3$, $C = -1$

1.4 Repeated factors

1. $A = 2$, $B = -1$, $C = 3$

2. $A = 3$, $B = 4$, $C = 2$

3. $A = 2$, $B = -3$

4. $A = 2$, $B = -1$, $C = -4$

5. $\dfrac{1}{x} - \dfrac{5}{x^2} + \dfrac{4}{2x-3}$

6. $\dfrac{4}{x} - \dfrac{3}{x-2} + \dfrac{5}{(x-2)^2}$

1.5 Algebraic division

1. $x - 2 + \dfrac{1}{x-3}$

2.

a. 15

b. $x - 6 + \dfrac{15}{x+1}$

3. $A = 1$, $B = -4$, $C = 12$, $D = -45$

4. $A = 1$, $B = 2$, $C = 8$, $D = 37$

5. $A = 6$, $B = -3$, $C = 1$

6. $A = 2$, $B = 2$, $C = 11$, $D = 32$

7. $A = 1$, $B = -5$, $C = 15$

8. $A = 2$, $B = -2$, $C = -3$

9. $A = 3$, $B = -5$, $C = 0$, $D = -2$, $E = 5$

Problem solving: Set A

Bronze **a** $\dfrac{3x-2}{x-1}$ **b** $\dfrac{3x+1}{x}$

Silver $\dfrac{x-3}{x-1}$

Gold $\dfrac{2}{3}x - \dfrac{1}{9} - \dfrac{52}{27x+18}$

Problem solving: Set B

Bronze $\dfrac{2}{3x-2} - \dfrac{1}{(3x-2)^2} + \dfrac{5}{1-x}$

Silver $x^2 - x - 6 = (x-3)(x+2)$. $A = 3$, $B = 5$, $C = -4$

Gold $\dfrac{1}{2(x+1)} - \dfrac{1}{2x-1} + \dfrac{3}{2(x-3)}$

Selected exam question bank answers

Question 5 $a = 2$, $b = 0$, $c = 11$, $d = -14$, $e = 9$

Question 9 $\dfrac{3}{x+1} - \dfrac{5}{(x+1)^2} + \dfrac{4}{x+2}$

Question 49

a. $A = 6$, $B = -2$, $C = 4$

b. $\ln \dfrac{784}{9}$

Question 62

a. Let $p = \sqrt{3}$ and $q = \dfrac{1}{\sqrt{3}}$; then $pq = \sqrt{3} \times \dfrac{1}{\sqrt{3}} = 1$
So if p and q are irrational numbers, it does not necessarily follow that pq is also an irrational number.

b. Assume that $\sqrt{2}$ is rational. Then $\sqrt{2} = \dfrac{a}{b}$, where a and b are integers, $b \neq 0$ and the highest common factor of a and b is 1. Then $a = \sqrt{2}b \Rightarrow a^2 = 2b^2$.
So a^2 is divisible by 2 and it must contain a (repeated) factor of 2, and so a is divisible by 2. So $a = 2p$, where p is an integer.
$\Rightarrow (2p)^2 = 2b^2 \Rightarrow b^2 = 2p^2$. So b^2 is divisible by 2 and so b is divisible by 2. As a and b are both divisible by 2, then the highest common factor of a and b is not 1. This contradiction implies that $\sqrt{2}$ is irrational.

Question 69

a. $$\frac{2(x-1)}{x^2-2x-8} - \frac{1}{x-4} = \frac{2(x-1)}{(x-4)(x+2)} - \frac{1}{x-4}$$
$$= \frac{2x-2-(x+2)}{(x-4)(x+2)} = \frac{x-4}{(x-4)(x+2)} = \frac{1}{x+2}$$

b. $0 < \mathrm{f}(x) < \dfrac{1}{6}$

c. $\mathrm{f}^{-1}(x) = \dfrac{1-2x}{x}$, domain $0 < x < \dfrac{1}{6}$

d. $x = 2\sqrt{5}$

Question 75

a. $\dfrac{x-5}{x-3}$

b. $1 - \dfrac{2}{x-3}$

c. $\mathrm{f}(x) = 1 - 2(x-3)^{-1}$

$\mathrm{f}'(x) = \dfrac{2}{(x-3)^2} > 0$ for all real values of x, $x > 3$

Therefore $\mathrm{f}(x)$ is an increasing function for all $x > 3$.

Question 78

a. $A = 3$, $B = -2$, $C = 4$

b. $-1 + x - \dfrac{5x^2}{2}$

c. $|x| < 1$